THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS CHAPTER TEST SET B

Chapter 11: **The World of Algorithms** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In the column-addition algorithm, the basic step we assume we can do directly is: 1

(a) adding two 10-digit numbers **(b)** counting dots one by one
(c) dividing by 10 **(d)** adding two single-digit numbers

2. When we move from adding 3-digit numbers to adding 5-digit numbers, which statement is correct? 1

(a) counting dots needs 2 times as many dots and the column method 100 times as many steps
(b) counting dots needs about 100 times as many dots, but the column method needs only 2 more columns
(c) both methods need about 100 times as much work
(d) both methods need only 2 more steps

3. In Euclid's algorithm the answer is reported as m when we reach gcd(m, n) with: 1

(a) m = 0 **(b)** m = n + 1 **(c)** n = 1 **(d)** n = 0

4. **Assertion (A):** The division algorithm of Āryabhaṭa needs fewer reductions than Euclid's subtraction algorithm for gcd(99, 2). 1

Reason (R): Repeated subtraction of 2 from 99 is replaced by one division giving the remainder.

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

5. While adding 999 and 1 by the column algorithm, the number of columns that produce a carry of 1 is: 1

(a) 5 **(b)** 2 **(c)** 4 **(d)** 3

6. By Euclid's algorithm, gcd(375, 825) is: 1

(a) 76 **(b)** 150 **(c)** 37 **(d)** 75

7. The number of times Step 3 of Euclid's subtraction algorithm is applied while computing gcd(825, 375) is: 1

(a) 7 **(b)** 8 **(c)** 9 **(d)** 6

8. Using the long-division (Āryabhaṭa) form of Euclid's algorithm, the number of reductions needed for gcd(60, 16) is: 1

(a) 2 **(b)** 3 **(c)** 5 **(d)** 4

9. The list of divisors of 36 produced by the divisor algorithm has how many entries? 1

(a) 8 **(b)** 11 **(c)** 9 **(d)** 10

10. The prime divisors of 90 (from divisors(90) = [1, 2, 3, 5, 6, 9, 10, 15, 18, 30, 45, 90]) are: 1

(a) 2, 3, 5 **(b)** 1, 2, 3, 5 **(c)** 2, 3, 5, 9 **(d)** 3, 5, 9

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. How many numbers must be checked to compute divisors(n) by the algorithm of the chapter? Why can the list of divisors be in increasing order? 2

12. Compute gcd(48, 18) by listing the common divisors. 2

13. What is the first improvement made to the gcd algorithm in Section 11.3.1? How many numbers need checking after it? 2

14. Why is it enough to check common divisors only up to min(m, n)? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe how the last improved gcd algorithm (using most-recent-common-divisor) works. Apply it to m = 8, n = 12. 3

16. Why does the work in the divisor-scan gcd algorithm grow 10 times when the smaller number goes from 3 digits to 4 digits? Why is this a drawback? 3

17. Prove: if d divides both m and n (m ≥ n), then d divides m − n. 3

OR

Prove: if d divides both n and m − n, then d divides m. 3

18. Compute gcd(42, 30) by Euclid's subtraction algorithm, writing each pair, and say how many times Step 3 was applied. 3

OR

Show that gcd(2k + 1, 2) needs many reductions in Euclid's subtraction algorithm but only 2 in the division algorithm. Take k = 4. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. State and prove the observation behind Euclid's algorithm: for m ≥ n, gcd(m, n) = gcd(n, m − n). 5

OR

Apply Euclid's subtraction algorithm to gcd(375, 825) showing every reduction (as the chapter does). State how many times Step 3 is used and why the algorithm is still slow for large inputs. 5

20. Explain Āryabhaṭa's division algorithm. Compute gcd(2587392, 157656) by this method and give all the divisions. 5

OR

Compute 5

(i) gcd(51000, 81000) by the improved Euclid algorithm, showing each division, and

(ii) gcd(1789287, 237656), stating the number of reductions.

END OF QUESTION PAPER — **Total: 40 Marks**